

# Failed Login Investigation

On 16 July 2026, Wazuh generated authentication alerts after five failed logon attempts were detected against the local Windows account "MrGreen." Investigation confirmed the activity originated from the local workstation due to repeated incorrect password entry. No indicators of automated brute-force activity or privilege escalation were identified. The incident was assessed as a benign authentication failure requiring no further escalation.

## Environment

- **SIEM:** Wazuh Manager (v4.14)
- **Endpoint:** Windows 11 workstation
- **Agent Status:** Active and reporting
- **Log Source:** Windows Security Event Logs (via Wazuh agent)
- **Monitoring:** Authentication events, failed logon attempts, account lockouts

## Scenario

A user enters the wrong password while attempting to authenticate to a Windows system. Wazuh detects multiple failed login attempts within a short time window, triggering authentication-related alerts.

## Alert Generated

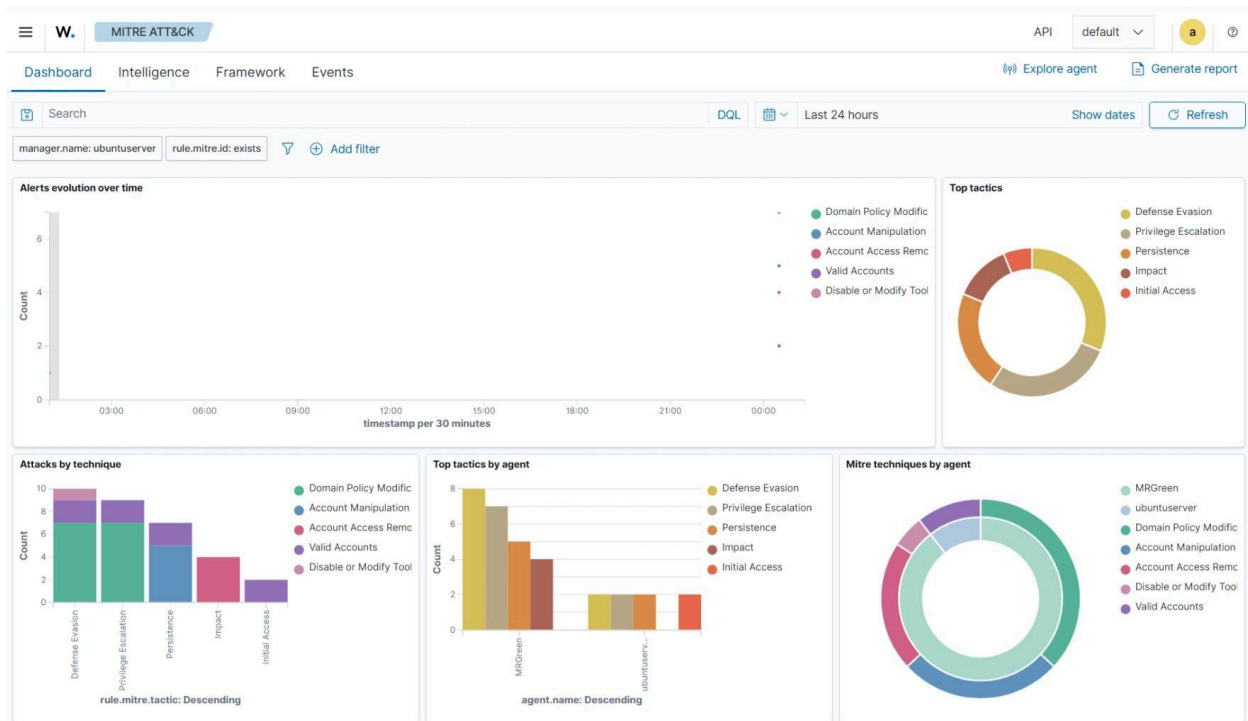
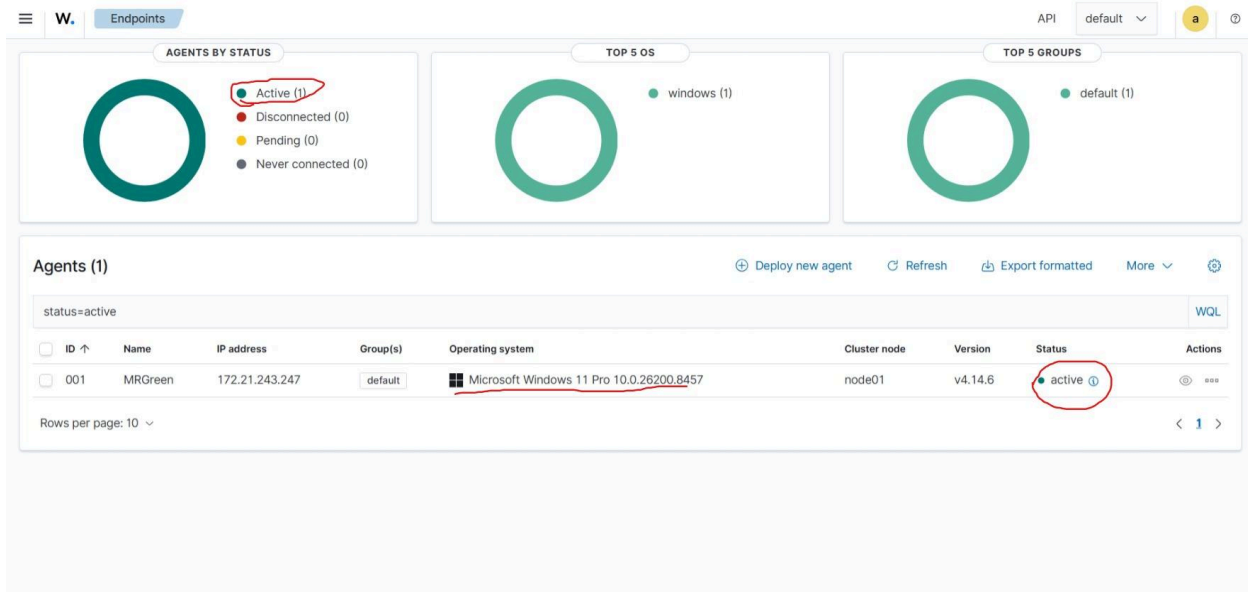
- **Rule Name:** Failed Windows Logon
- **Rule ID:** 60122
- **Alert Level:** 5
- **Detection Source:** Windows Security Logs
- **Event IDs Observed:**
  - 4625 – Failed Logon Attempt
- **Trigger Pattern:** Multiple failed logons within 1 minute

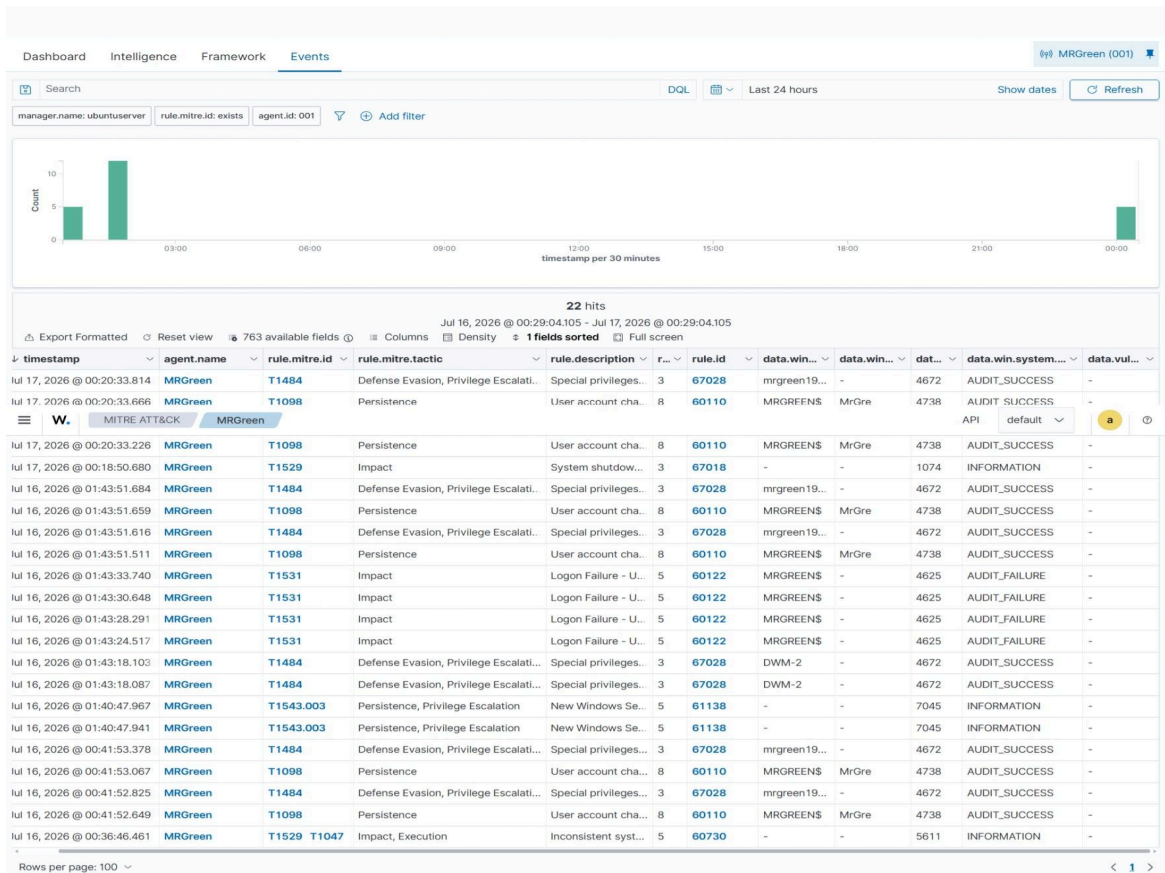
## Evidence

### 1. Wazuh Dashboard

- Alert entries showing repeated failed logon attempts
- Timestamp of each alert
- Source IP

- Username involved





## 2. Windows Event Logs

From **Event Viewer** → **Security Logs**:

*Event ID: 4625*  
*Account Name: MrGreen*  
*Failure Reason: Unknown username or bad password*  
*Logon Type: 2 (Interactive)*  
*Source Address: 127.0.0.1*

17 hits  
Jul 12, 2026 @ 00:00:00.000 - Jul 18, 2026 @ 23:59:59.999

Export Formatted Reset view 763 available fields Columns Density 1 fields sorted Full screen

| timestamp                   | agent.name | rule.mitre.id | rule.mitre... | rule.description       | rule.level | rule.id | data... | data.win.ev... | data.win.eventdata... | data.win.system.se |
|-----------------------------|------------|---------------|---------------|------------------------|------------|---------|---------|----------------|-----------------------|--------------------|
| Jul 16, 2026 @ 01:43:33.740 | MRGreen    | T1531         | Impact        | Logon Failure - Unk... | 5          | 60122   | 4625    | 127.0.0.1      | MRGREEN\$             | AUDIT_FAILURE      |
| Jul 16, 2026 @ 01:43:30.648 | MRGreen    | T1531         | Impact        | Logon Failure - Unk... | 5          | 60122   | 4625    | 127.0.0.1      | MRGREEN\$             | AUDIT_FAILURE      |
| Jul 16, 2026 @ 01:43:28.291 | MRGreen    | T1531         | Impact        | Logon Failure - Unk... | 5          | 60122   | 4625    | 127.0.0.1      | MRGREEN\$             | AUDIT_FAILURE      |
| Jul 16, 2026 @ 01:43:24.517 | MRGreen    | T1531         | Impact        | Logon Failure - Unk... | 5          | 60122   | 4625    | 127.0.0.1      | MRGREEN\$             | AUDIT_FAILURE      |

## 3. Additional Context

- Number of failed attempts
- Whether the account became locked

- Whether attempts originated locally or remotely

## Investigation

1. **Verified log ingestion** Confirmed that Windows Security logs were successfully collected by the Wazuh agent and visible in the Wazuh dashboard.
2. **Reviewed Wazuh alerts** Identified multiple alerts tied to Event ID 4625, indicating failed authentication attempts.
3. **Correlated timestamps** Matched Wazuh alert timestamps with Windows Event Viewer logs to ensure consistency.
4. **Analyzed logon details** Examined logon type, failure reason, and source network address to determine whether attempts were local or remote.
5. **Identified user and source** Determined that the failed attempts were associated with the user *MrGreen* from *local machine*.
6. **Assessed intent** Based on frequency and pattern, determined whether the activity was accidental (user mistyping password).

## Severity Assessment

Incident Severity

Low

Reason

Single user

Local workstation

Known user

No persistence

No privilege escalation

No malware

No lateral movement

## Findings

- Windows logs were successfully collected and forwarded to Wazuh.
- Wazuh correctly generated alerts for repeated failed logon attempts.
- Event ID **4625** was consistently logged for each failed attempt.
- The attempts originated from a local *machine*.
- The user *Mr.Green* entered incorrect credentials **5** times within **1** minute.
- No evidence of brute force automation unless the pattern suggests otherwise.
- No lateral movement or privilege escalation detected.

## Root Cause

- **Repeated incorrect password entry by the legitimate user caused Windows Event ID 4625 events, which Wazuh correctly detected and alerted on.**

## MITRE ATT&CK Mapping

Although the generated events match those commonly associated with ATT&CK technique T1110 (Brute Force), this investigation concluded the activity resulted from legitimate user error rather than malicious password guessing. Therefore, the ATT&CK mapping is included for detection reference rather than adversary attribution.

## Recommendations

- **Enable account lockout policy** to limit repeated failed attempts.
- **Educate users** on proper password entry and avoiding repeated attempts.
- **Monitor for abnormal patterns**, such as attempts across multiple accounts.
- **Enable Sysmon** for deeper authentication telemetry.
- **Configure Wazuh thresholds** to alert earlier on suspicious authentication patterns.

## Lessons Learned

- Wazuh effectively detects authentication anomalies when Windows logs are properly ingested.
- Failed login investigations rely heavily on correlating Event IDs with SIEM alerts.
- Clear documentation helps build repeatable workflows for future incidents.
- Authentication failures can be benign or malicious , context and frequency matter.

This investigation demonstrated how repeated authentication failures appear in both Windows Event Logs and Wazuh alerts. It reinforced the importance of correlating multiple data sources before determining whether authentication anomalies represent malicious activity or normal user behavior. Future investigations will include Sysmon telemetry and additional endpoint context to improve detection confidence.